

Topic 5: Communication and the internet

Computer networks and the internet are now ubiquitous. Many computer applications in use today would not be possible without networks. Students should understand the key principles behind the organisation of computer networks. Ideally, they should be able to experiment by setting up a simple network. Our increasing reliance on computer networks makes us vulnerable to a range of attacks from cyber criminals. As IT system security defences become more robust, attack methods become more sophisticated. Students should have a good understanding of the current IT security threats and how to apply appropriate protection methods.

	Students should:
5.1 Networks	5.1.1 Understand why computers are connected in a network. 5.1.2 Understand the different types of networks (local area network (LAN), wide area network (WAN), personal area network (PAN)) and usage models (client-server, peer-to-peer). 5.1.3 Understand wired and wireless connectivity. 5.1.4 Understand that network data speeds are measured in bits per second (Mbps, Gbps). 5.1.5 Understand the role of and need for network protocols (Ethernet®, Wi-Fi, TCP/IP, HTTP, HTTPS, FTP, email (POP3, SMTP, IMAP)). 5.1.6 Understand that data can be transmitted in packets using layered protocol stacks and the 4-layer TCP/IP model (application, transport, network, data link). 5.1.7 Understand characteristics of network topologies (bus, ring, star, mesh). 5.1.8 Understand the different mobile communication standards (3G, 4G and subsequent generations).